| | |
|---|---|
| | **DES Pune University, Pune** |
| Academic Year: 2026-27 | |
| Semester: I (Odd Semester) | **School of Engineering and Technology** |
| Class & Div: TY ECE - B | **Electronics and Communication Engineering (AI & ML)** |
| **PBEL Logbook Record** | |

| S. No. | Date (From-To) | Planned Activities | Work Carried Out | Tools / Techniques Used | Observations / Results | Issues / Challenges | Action Plan for Next Week | Student Signature | Mentor Remarks & Signature |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 06/08/2026 to 06/08/2026 | Meet the allotted mentor, understand the PBEL guidelines and evaluation scheme, and discuss possible project ideas in agriculture using IoT and AI. | The mentor was allotted on 05/08/2026. Met the mentor for the first time and discussed the PBEL guidelines, the report/proposal formats and the logbook requirements. Brainstormed problems in agriculture such as manual irrigation, water wastage and late detection of crop diseases. Shortlisted the idea of a Smart Rice Farming System with two parts: (1) sensor-based field monitoring with pump control and (2) AI-based rice disease detection from leaf images. The mentor advised keeping a scope that can be completed in one semester with easily available hardware. | Discussion with mentor, PBEL guidelines and formats, brainstorming, internet search. | The project topic was shortlisted and accepted by the mentor in principle. Two modules were identified: IoT monitoring and irrigation, and AI-based disease detection. The team divided the initial tasks (literature survey, hardware list, dataset search) among the four members. | Deciding a scope that is achievable within the semester and with the available hardware and time. Choosing which rice diseases to include. | Start the literature survey on IoT irrigation and rice disease detection, prepare a list of sensors and components, and decide the three diseases to detect. | | |
| 2 | 19/08/2026 to 19/08/2026 | Present the literature survey and finalise the problem statement, objectives and component list for mentor review. | Studied research papers on IoT-based smart irrigation systems, soil-moisture and rain sensing, and YOLO-based plant disease detection. Prepared the problem statement and objectives. Selected Rice Blast, Rice Blight and Brown Spot as the diseases to detect. Prepared the component list: soil moisture sensor, rain sensor, DHT11, LM35, LDR, relay module and water pump, with a microcontroller (ESP8266 initially planned). The mentor reviewed the draft and suggested simplifying the objectives. | IEEE Xplore, SpringerLink, MDPI, ScienceDirect, MS Word, discussion with mentor. | Problem statement and objectives were finalised. Existing work was found to cover either irrigation or disease detection; the combined approach for rice was confirmed as the project direction. Reference list of research papers started. | Few papers combine irrigation monitoring and disease detection for rice. Understanding the technical details of YOLO from research papers took time. | Write the project proposal in the PBEL proposal format and arrange the hardware components. | | |

Project Title: Smart Rice Farming System
Project Mentor: Dr. Hrishikesh Vanjari
Project Reviewer: Dr. Madan Mali

| | |
|---|---|
| | **DES Pune University, Pune** |
| Academic Year: 2026-27 | |
| Semester: I (Odd Semester) | **School of Engineering and Technology** |
| Class & Div: TY ECE - B | **Electronics and Communication Engineering (AI & ML)** |
| **PBEL Logbook Record** | |

| S. No. | Date (From-To) | Planned Activities | Work Carried Out | Tools / Techniques Used | Observations / Results | Issues / Challenges | Action Plan for Next Week | Student Signature | Mentor Remarks & Signature |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 01/09/2026 to 01/09/2026 | Submit the proposal draft for mentor approval and start collecting components and rice disease datasets. | Prepared the proposal with abstract, introduction, problem statement, objectives, brief literature review, proposed methodology, applications and references. The mentor reviewed it and suggested corrections in the methodology section. Collected the sensors and the relay module from the laboratory and local market. Searched for publicly available rice-leaf disease image datasets and checked their classes and image quality. | MS Word, PBEL proposal format, online dataset repositories, discussion with mentor. | The proposal was approved after minor corrections. Most hardware components became available. Candidate datasets containing Blast, Blight and Brown Spot images were identified. | Some components were not available locally and had to be arranged separately. Dataset images differ in quality, background and labelling style. | Interface all sensors with the microcontroller and check the readings on the serial monitor. | | |
| 4 | 10/09/2026 to 10/09/2026 | Interface all the sensors with the microcontroller and verify their readings. | Connected the soil moisture, rain, DHT11, LM35 and LDR sensors and printed their values on the serial monitor. Found that the ESP8266 has only one analog input, which is not enough for the four analog sensors (soil moisture, rain, LM35, LDR). The design was therefore changed to the ESP32, which has multiple 12-bit ADC channels and more GPIO pins. Re-wired the sensors to the ESP32 and tested each sensor separately in dry and wet conditions. | Arduino IDE, ESP32 development board, sensors, breadboard, jumper wires, multimeter. | All sensor values were read successfully using the ESP32. The soil moisture value changed clearly between dry soil and wet soil, and the rain sensor responded to water drops. The change from ESP8266 to ESP32 was noted for the report. | Raw analog readings fluctuate and need calibration and averaging. Low-cost sensors such as the DHT11 have limited accuracy. | Create an Adafruit IO account, set up the feeds and send the sensor data to the cloud over Wi-Fi. | | |

Project Title: Smart Rice Farming System
Project Mentor: Dr. Hrishikesh Vanjari
Project Reviewer: Dr. Madan Mali

| | |
|---|---|
| | **DES Pune University, Pune** |
| Academic Year: 2026-27 | |
| Semester: I (Odd Semester) | **School of Engineering and Technology** |
| Class & Div: TY ECE - B | **Electronics and Communication Engineering (AI & ML)** |
| **PBEL Logbook Record** | |

| S. No. | Date (From-To) | Planned Activities | Work Carried Out | Tools / Techniques Used | Observations / Results | Issues / Challenges | Action Plan for Next Week | Student Signature | Mentor Remarks & Signature |
|---|---|---|---|---|---|---|---|---|---|
| 5 | 23/09/2026 to 23/09/2026 | Connect the ESP32 to Adafruit IO and display the sensor data on a dashboard. | Created an Adafruit IO account and made a separate feed for each parameter. Wrote the ESP32 code to connect to Wi-Fi and Adafruit IO and to publish soil moisture, rain, temperature, humidity and light values at regular intervals. Converted raw readings to useful units (soil moisture in percent, LM35 in degrees Celsius). Built an Adafruit IO dashboard with gauges and line charts for the feeds. | Adafruit IO, Adafruit IO Arduino library, DHT sensor library, Arduino IDE, Wi-Fi hotspot, serial monitor. | Live sensor values appear on the Adafruit IO dashboard and the history is stored in the feeds, which can later be used as data for an ML model. The values on the dashboard matched the serial monitor. | Occasional Wi-Fi disconnection during testing. Adafruit IO limits the data rate, so the upload interval had to be chosen carefully. | Connect the relay module and pump to the ESP32 and control the pump from the dashboard. | | |
| 6 | 01/10/2026 to 01/10/2026 | Implement remote pump control from the Adafruit IO dashboard and test it. | Connected the relay module and the water pump to the ESP32. Created a "pump" feed and added a toggle block on the dashboard. Wrote the code so that the ESP32 subscribes to the pump feed and switches the relay ON or OFF when the user changes the toggle. Kept the pump OFF at start-up for safety and tested ON/OFF switching several times, also after restarting the board. | ESP32, relay module, DC water pump, external power supply, Adafruit IO toggle block, Arduino IDE. | The pump turns ON and OFF according to the dashboard toggle. Sensor upload and pump control work together on the same ESP32. The control flow (dashboard to feed to ESP32 to relay to pump) was documented for the report. | The pump needed a separate supply and a common ground with the ESP32. Pump control needs internet access and a user decision; it is not automatic. | Collect and label rice-leaf images for Rice Blast, Rice Blight and Brown Spot. | | |

Project Title: Smart Rice Farming System
Project Mentor: Dr. Hrishikesh Vanjari
Project Reviewer: Dr. Madan Mali

| | |
|---|---|
| | **DES Pune University, Pune** |
| Academic Year: 2026-27 | |
| Semester: I (Odd Semester) | **School of Engineering and Technology** |
| Class & Div: TY ECE - B | **Electronics and Communication Engineering (AI & ML)** |
| **PBEL Logbook Record** | |

| S. No. | Date (From-To) | Planned Activities | Work Carried Out | Tools / Techniques Used | Observations / Results | Issues / Challenges | Action Plan for Next Week | Student Signature | Mentor Remarks & Signature |
|---|---|---|---|---|---|---|---|---|---|
| 7 | 08/10/2026 to 08/10/2026 | Prepare the rice disease image dataset for model training. | Collected rice-leaf images from the available datasets and checked them for quality. Manually labelled the diseased regions for the three classes Rice Blast, Rice Blight and Brown Spot using an annotation tool and saved the labels in YOLO format. Divided the labelled images into training, validation and test sets and created the dataset configuration file for training. | Image annotation tool, Python, available rice disease datasets, file organisation scripts. | A labelled dataset with three classes was prepared and split for training, validation and testing. The class-wise image counts were noted for the report. | Manual labelling is time-consuming. The number of images differs between classes, which can affect model accuracy. | Train the YOLO model on the labelled dataset and evaluate it. | | |
| 8 | 14/10/2026 to 14/10/2026 | Train the YOLO model and check its detection results on test images. | Trained a YOLO object-detection model using the Ultralytics framework on the labelled rice-leaf dataset, starting from pre-trained weights. Monitored the training and validation curves. Evaluated the model on the validation and test images using precision, recall and mAP, studied the confusion matrix and tested the detection on sample leaf images. | Python, Ultralytics YOLO, PyTorch, Google Colab / GPU system, matplotlib. | The trained model detects the three diseases and gives a disease class with a confidence value for each detection. The evaluation results (precision, recall, mAP) and the detection images were saved for the project report. | Some detections have low confidence, and some classes are confused with each other. Performance on real field images may differ from the dataset images. | Test both subsystems once more, collect screenshots and photographs, and write the final report. | | |

Project Title: Smart Rice Farming System
Project Mentor: Dr. Hrishikesh Vanjari
Project Reviewer: Dr. Madan Mali

| | |
|---|---|
| | **DES Pune University, Pune** |
| Academic Year: 2026-27 | |
| Semester: I (Odd Semester) | **School of Engineering and Technology** |
| Class & Div: TY ECE - B | **Electronics and Communication Engineering (AI & ML)** |
| **PBEL Logbook Record** | |

| S. No. | Date (From-To) | Planned Activities | Work Carried Out | Tools / Techniques Used | Observations / Results | Issues / Challenges | Action Plan for Next Week | Student Signature | Mentor Remarks & Signature |
|---|---|---|---|---|---|---|---|---|---|
| 9 | 19/10/2026 to 19/10/2026 | Show the working prototype and the trained model to the mentor and discuss the final report. | Demonstrated the ESP32 sensor data upload to Adafruit IO, the pump control from the dashboard and the rice disease detection output. Discussed the changes from the proposal: ESP32 instead of ESP8266, Adafruit IO as the dashboard, and user-controlled pump instead of automatic irrigation. Prepared the report in the PBEL format with results, future enhancement and conclusion sections. | ESP32 prototype, Adafruit IO dashboard, trained YOLO model, MS Word, PBEL report format. | Both subsystems worked as expected. The mentor suggested adding screenshots, result tables and the future enhancements: training an ML model on the collected IoT data and merging the AI model with the IoT dashboard. | The report still needs result figures, screenshots and the plagiarism check. The AI and IoT parts are still separate systems. | Complete the report, run the plagiarism check, complete the logbook and obtain mentor and reviewer signatures. | | |
| 10 | 23/10/2026 to 23/10/2026 | Submit the final PBEL project report, proposal, logbook and supporting records after mentor and reviewer verification. | Checked the final versions of the project report, proposal and logbook, and verified that the report follows the PBEL format. Obtained the required signatures, printed the documents and prepared the complete set with the supporting records (plagiarism report, source code, trained model and dataset) for submission. | MS Word, PDF export, plagiarism checker, submission checklist, printing, digital backups. | All required documents verified and signed by the mentor and the reviewer, ready for submission before the deadline. Submission acknowledgement will be recorded after handover. | Missing signatures or mismatched document versions could delay acceptance. All documents must be complete and consistent. | Record the final submission date and acknowledgement. Keep a backup of the report, source code, trained model and dataset. | | |

Project Title: Smart Rice Farming System
Project Mentor: Dr. Hrishikesh Vanjari
Project Reviewer: Dr. Madan Mali